Amin Mamandipoor

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EDUCATION

University of Kansas, PhD in Computer Science — 2022 - May 2027 (expected)
University of Tabriz, MSc in Computer Engineering — 2018 - 2021
University of Tabriz, BSc in Computer Engineering — 2013 - 2018

PROFESSIONAL EXPERIENCE

Clockwork, Software Engineer Intern | May 2026 - Aug 2026 | USA

- Built and deployed backend services for orchestrating and migrating large-scale distributed AI training workloads across Kubernetes and SLURM GPU clusters on public cloud infrastructure.
- Improved reliability and portability of production training jobs by hardening checkpoint/restore and job-migration paths and validating failure and edge cases across cluster configurations.
- Worked on network observability, CI/CD, high-performance RDMA networking, and network interface cards (NICs).

University of Kansas, Graduate Research Assistant | Jan 2022 - Present | Lawrence, KS

- **SoC memory architecture & Quality of Service (QoS):** Designed a per-bank bandwidth regulator on a RISC-V SoC for shared-resource management and memory isolation; blocked a DRAM-contention attack causing up to 81x throughput loss and improved best-effort throughput 5.7x (RTAS 2026).
- **Performance modeling & cache architecture:** Developed SMT support and cache-banking models in gem5 full-system simulation; modified the Linux kernel for deterministic memory access; co-authored a gem5 profiling study (ISPASS 2023).
- **Architecture trade-offs (SDT):** Architected per-core simultaneous data-delivery threads to isolate network processing from applications; matched a 2x-larger chip's network throughput within 10% using 47.5% less area and 66% less power (IEEE CAL 2025).
- **Multi-accelerator systems (DMX):** Co-designed programmable data-restructuring engines with UC San Diego and IBM Research, chaining heterogeneous accelerators without CPU mediation; up to 8.2x lower latency, 13.6x higher throughput, 5.2x energy savings (HPCA 2024).
- **Memory-system architecture (SmartDIMM):** Prototyped in-DIMM encryption and compression on Samsung's AxDIMM while preserving the DRAM and DDR interfaces; achieved up to 10.3x higher requests/s and 36-89% lower memory bandwidth than CPU and SmartNIC offload (HPCA 2024).
- **Far memory (XFM):** Hid compression traffic inside DRAM refresh windows, removing swap-induced memory interference and improving application performance 5-27% (MICRO 2023).

PUBLICATIONS

SBA: A Practical Memory Performance Denial-of-Service Attack
A. Mamandipoor, C. Sullivan, C. Strickler, M. Alian, H. Yun — uASC | 2027

Per-Bank Memory Bandwidth Regulation for Predictable and Performant Real-Time Systems
C. Sullivan, **A. Mamandipoor**, C. Strickler, H. Yun — IEEE RTAS | 2026

SDT: Cutting Datacenter Tax Through Simultaneous Data-Delivery Threads
A. Mamandipoor, H. Tran, M. Alian — IEEE CAL | 2025

DMX: Chaining Cross-Domain Multi Accelerators
S. Wang, H. Xu, **A. Mamandipoor**, R. Mahapatra, B. Ahn, S. Ghodrati, K. Kailas, M. Alian, H. Esmaeilzadeh — IEEE HPCA | 2024

SmartDIMM: In-Memory Acceleration of Upper Layer Protocols
N. Patel, **A. Mamandipoor**, M. Nouri, M. Alian — IEEE HPCA | 2024

XFM: Accelerated Software-Defined Far Memory
N. Patel, **A. Mamandipoor**, D. Quinn, M. Alian — IEEE/ACM MICRO | 2023

Profiling gem5 Simulator
J. Umeike, N. Patel, A. Manley, **A. Mamandipoor**, H. Yun, M. Alian — IEEE ISPASS | 2023

TEACHING EXPERIENCE

University of Kansas, Graduate Teaching Assistant | Jan 2024 - Present | Lawrence, KS

- Senior Design (EECS 541/542), Embedded Systems (EECS 388), and Computer Architecture (EECS 645): mentored project teams and led labs and tutorials on cache hierarchies, gem5, interrupts, and MIPS assembly.

AWARDS

Best Student Paper Award, IEEE RTAS (2026) | **First Place**, I2S Student Research Symposium (2024) | **Second Place**, Samsung Open Innovation Contest for AxDIMM Technology (2022) | **Travel Grants**: gem5 Bootcamp (2024), ISPASS (2023), MICRO (2022)

SKILLS

Architecture & Systems: RISC-V SoCs, cache modeling, DRAM, memory QoS, shared-resource management, memory isolation, performance modeling.

Programming: C, C++, CUDA, Python, Java, Bash/Shell, ARM assembly

HDLs: Verilog, SystemVerilog, SystemC, VHDL

Architecture Tools: gem5, FireSim, ModelSim, Cadence Virtuoso, Vivado HLS, Vitis AI

Performance Analysis: Linux perf, Intel VTune, Intel PIN, Intel CAT/MBA, valgrind, GDB, LLM benchmarking (llama.cpp)